Certified Weil Positivity at $L = 4/5$: Two-Sided Bounds, Parity Separation, and Cutoff Diagnostics

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November 2026 — Technical Note (preprint), version 1.0.2

Abstract

For the Weil explicit-formula quadratic form restricted to test functions supported in a compact window $[-L, L]$ with $L = 4/5$, this note reports a computer-assisted two-sided bound on its spectral infimum $\lambda_Q(4/5)$ in one fixed normalization:

$$1.41 \times 10^{-17} \;\le\; \lambda_Q(4/5) \;\le\; 1.9014 \times 10^{-17}.$$

The lower bound is obtained by a frequency-cutoff comparison at $T = 400$, reduced to a 440-dimensional Legendre block whose positive definiteness is verified by a preconditioned congruence in Arb ball arithmetic, together with a super-exponentially small complement estimate and a Schur-type coupling bound; both parity sectors are certified. The upper bound is the Rayleigh quotient of the *original* form at one explicit even rational test function, evaluated in the time domain through the autocorrelation and a compact archimedean identity, so no oscillatory frequency tail is integrated. A separate certificate gives $\lambda_{Q,\mathrm{odd}} \ge 2.3 \times 10^{-17}$, which exceeds the upper bound and therefore shows that the even infimum determines λ_Q. Diagnostics at cutoffs $T \in \{200, 400, 800\}$ measure the discarded exterior contribution and bound the gain available from a larger cutoff. All constants are published as midpoint–radius enclosures with a cold verifier. This is a single-window computation: it establishes no arithmetic estimate, nothing about the location of any zeta zero, and nothing about the Riemann hypothesis.

1 Introduction

1.1 The object

Weil's explicit formula pairs a test function against the zeros of ζ, the pole at $s = 1$, the archimedean place and the primes. When the test function is required to be supported in a fixed compact window, the resulting quadratic form becomes a concrete object: a bounded-below Hermitian form on $L^2(-L, L)$ whose positivity *for every* L is equivalent to the Riemann hypothesis [8, 1], and whose positivity at one fixed L is not. Finite-window Weil positivity in this sense is the object studied in the spectral programs of Connes and van Suijlekom [4], Connes and Consani [2], and Connes, Consani and Moscovici [3], and quantitatively in Zhu [9], which supplies the finite-reduction strategy that the present computation follows.

The quantity computed here is the spectral infimum

$$\lambda_Q(L) \;=\; \inf_f \frac{Q_{\zeta,L}(f)}{\|f\|_2^2}$$

of that form at $L = 4/5$, in the normalization fixed in Section 2. Two things make a two-sided statement possible at this window and difficult at larger ones. An *upper* bound needs a single admissible test function and is unconditional. A *lower* bound needs the whole window, hence a reduction of an infinite-dimensional positivity question to a finite matrix plus certified tails; the cost of that reduction grows very fast in L, which is the content of Theorem 1.4 of [9].

1.2 What this note reports

1. A certified lower bound $Q(f) \geq 1.41 \times 10^{-17} \|f\|_2^2$ at $L = 4/5$, in both parity sectors and hence for arbitrary complex tests (Section 3).

2. A certified upper bound $\lambda_Q(4/5) \leq 1.9014 \times 10^{-17}$ from one explicit even rational test function, with the *original* form evaluated in the time domain (Section 4).

3. A certified parity separation: the odd sector satisfies $\lambda_{Q,\mathrm{odd}} \geq 2.3 \times 10^{-17}$, which is strictly above the upper bound, so the even infimum determines λ_Q (Section 5).

4. Cutoff diagnostics at $T \in \{200, 400, 800\}$ that measure what the comparison discards on fixed test functions, and a certified ceiling on what a larger cutoff can deliver (Section 6).

1.3 Relation to prior work, and what is not claimed

The finite-to-infinite reduction strategy — a scalar gate on the exterior of a frequency cutoff, a Legendre/Bessel representation of the interior, and super-exponentially small tails — is Theorem 1.1 of [9]. Zhu describes supplementary source code and certificates in [9, §17]; these were not obtained or executed in the present work, so his matrix step is neither re-run nor verified here. The reduction was instead rederived in the normalization of the present repository, and every constant below was computed from scratch. The bracket $8.9 \times 10^{-18} \leq \lambda_{\min}(0.8) \leq 2.27 \times 10^{-17}$ stated in the abstract of [9] was used only as an external comparison point and never as an input, an expected answer, or a tolerance. The parity structure is likewise not new, and at this window it is not merely the decoupling that is already published: [9, Thm. 6.2] certifies the *separation* at support 1.6 — that is, at $L = 4/5$ — and does so with a far stronger odd-sector constant than the one obtained here, concluding in addition that the ground state is simple and even. What Section 5 contributes is an independent re-derivation of the ordering from this pipeline's own certified data, nothing structural.

The computation continues the implementation developed alongside our finite-window sensitivity study [10], whose repository supplies the zeta-window normalization of Section 2, the geometric-side assembly, the pole conventions, and the closed-form archimedean evaluator used as the validation reference in Section 4(a). The even/odd pole signs used here are the ones checked entrywise against that assembly in our real-zero parity study [11]. Neither the δ^2–L^3 sensitivity estimate of [10] nor the planted-real-zero analysis of [11] is an input to the lower bound proved here: the first concerns a counterfactual perturbation of the zero set, and the second a configuration that is nowhere assumed to exist. The finite-to-infinite reduction follows [9]; what is contributed here is its execution in certified arithmetic, together with the resulting bracket, the numerical parity separation of Section 5, and the cutoff diagnostics of Section 6.

What follows is deliberately narrow, and the following are *not* claimed. No arithmetic estimate is obtained: nothing here improves any bound on primes, on $\Lambda(n)$, on $\psi(x)$, or on a zero-free region. No statement is made about the location of any zero of ζ, and no range of zeros is verified. No new positivity framework is introduced; the framework is that of the cited sources, and the contribution is a certified computation inside it. The computation is at one window, $L = 4/5$, and gives nothing at any larger support — there is no propagation theorem here, and a certificate at one L does not provide one. Nothing here bears on the Riemann hypothesis. This

is a technical note, not peer-reviewed research, and it is not claimed that anyone else uses or has used this work.

1.4 The same window, compared

Because [9] certifies at support 1.6, its results and the present ones describe the same object in the same normalization (its equations (2)–(3) are (1)–(2) below), so they can be put side by side. Doing so is the fairest summary of what this note adds and what it does not.

quantity at $L = 4/5$	[9], certified	this note
whole-window infimum	$[8.9 \times 10^{-18},\ 2.27 \times 10^{-17}]$	$[1.41 \times 10^{-17},\ 1.9014 \times 10^{-17}]$
odd-sector lower bound	8.206×10^{-15}	2.3×10^{-17}
second even min–max value	$\geq 2.085 \times 10^{-12}$	not certified here
simplicity of the ground state	[9, Thm. 6.2]	not independently established here

The one place this note improves on the published result is the width of the whole-window bracket, which falls from 1.38×10^{-17} to 4.914×10^{-18}: approximately 64.4% narrower, a reduction by a factor of 2.81. The present bracket is strictly nested inside the published one, both endpoints moving inward. In the other three rows [9] is stronger, and in the odd sector by more than two orders of magnitude.

As a consistency check — not as an endpoint, and not as a replay of his certificate — [9] also reports a numerical reference estimate $\lambda_{\min}(Q|_{\text{window}}) = 1.656 \times 10^{-17}$ for this window, from his own exact time-domain assembly. That value lies inside the bracket certified here. It is a non-certified reference value in [9] and is used here only in that capacity.

2 Setup, conventions, and the statement

2.1 Normalization

Fix $L > 0$. For $f \in L^2(-L, L)$, extended by zero, write

$$F(t) \;=\; \int_{-L}^{L} f(u)\, e^{itu}\, du, \qquad \text{so} \qquad \int_{\mathbb{R}} |F(t)|^2\, dt \;=\; 2\pi \|f\|_2^2.$$

The window Weil form used throughout is

$$Q(f) \;=\; 2s\, F(i/2)^2 \;+\; \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2\, \Psi_L(t)\, dt, \tag{1}$$

$$\Psi_L(t) \;=\; H(t) - P_L(t), \qquad H(t) \;=\; \operatorname{Re} \psi\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big) - \log \pi, \qquad P_L(t) \;=\; \sum_{\log n < 2L} \frac{2\Lambda(n)}{\sqrt{n}} \cos(t \log n). \tag{2}$$

Here ψ is the digamma function and Λ is the von Mangoldt function. The pole sign is $s = +1$ on the even sector and $s = -1$ on the odd sector; the pole term is $2F(i/2)F(-i/2) = 2(C^2 - S^2)$ with $C = \int f \cosh(u/2)$ and $S = \int f \sinh(u/2)$, so C survives on even f and S on odd f, which is exactly what the two signs encode. These signs are not asserted here: in the parity study [11, §3(c)] the $\delta = 1/2$ real-pair perturbation is compared entrywise with this repository's own pole term $2\,\text{pole_sign}(\varepsilon) P_j P_k$ and agrees in all nine tested cells to 6.22×10^{-61}, with pole_sign $= -1$ on the odd sector, while the opposite sign convention is wrong by $O(1)$ to $O(10)$; and [11, Remark 6] identifies that $\delta = 1/2$ case as the parity splitting of [9, Lemma 6.1]. The prime sum in (2) is *finite and exact*: the autocorrelation of a function supported in $[-L, L]$ is supported

in $[-2L, 2L]$, so only n with $\log n < 2L$ contribute and there is no truncation. At $L = 4/5$ the active prime powers are $\{2, 3, 4\}$.

This is the normalization of `src/zeta_window.py` in the repository accompanying this note, documented in [10], where the same object is written $T_R = A - 2M$ with $A = \text{Pole} + \text{Arch} - (\log \pi)I$ and $2M = \text{Prime}$. It is an imported convention, not derived here, and the grouping of $-\log \pi$ is part of it. It was cross-checked against the repository's own independent closed form for the constant test function (Section 4, item (a)).

2.2 The admissible class

Since H is even, continuous, and increasing on $t > 0$ (Lemma 5), and $|P_L| \le A_L := \sum_{\log n < 2L} 2\Lambda(n)/\sqrt{n} < \infty$, the function Ψ_L is bounded below on $\mathbb{R}$ and bounded on compacts. Hence for *every* $f \in L^2(-L, L)$ the negative part of the integral in (1) is finite and $Q(f)$ is well defined in $(-\infty, +\infty]$. We therefore take the admissible class to be all of $L^2(-L, L)$, with the convention that a lower bound $Q(f) \ge \mu \|f\|_2^2$ is satisfied trivially where $Q(f) = +\infty$. This is a choice of convention made in this note; it strengthens rather than weakens the statement, and it removes any need for a density or form-domain argument.

2.3 Three spectral quantities, kept apart

Confusing the following three numbers is the easiest way to overstate a result of this kind, so they are named separately and used separately throughout.

Definition 1. With $T > 0$ a frequency cutoff, β a constant with $\Psi_L(t) \ge \beta$ for $|t| \ge T$, and V_N a finite-dimensional subspace of $L^2(-L, L)$:

$$\begin{aligned} \lambda_Q &= \inf\{\, Q(f)/\|f\|_2^2 \,\} && \text{over the admissible class,} \\ \lambda_R(T) &= \inf\{\, R_{T,\beta}(f)/\|f\|_2^2 \,\} && \text{over the same class, } R_{T,\beta} \text{ as in (3),} \\ \lambda_A(T, N, \varepsilon) &= \lambda_{\min}(\text{compression of } R_{T,\beta} \text{ to } V_N) && \text{in the parity sector } \varepsilon. \end{aligned}$$

Because $Q \ge R_{T,\beta}$ pointwise, and because an infimum over a subspace is at least the infimum over the whole space,

$$\lambda_R(T) \;\le\; \lambda_Q, \qquad \lambda_R(T) \;\le\; \lambda_A(T, N, \varepsilon),$$

and there is **no** inequality in either direction between λ_A and λ_Q. In particular a numerical least eigenvalue of a finite block is not an upper bound for λ_Q, and only the evaluation of Q itself on an explicit test function is.

2.4 The theorem

Theorem 2. *Let $L = 4/5$ and let Q be the form (1)–(2) on $L^2(-L, L)$.*

(i) (Lower bound.) *For every $f \in L^2(-4/5, 4/5)$,*

$$Q(f) \;\ge\; \tfrac{141}{10^{19}}\, \|f\|_2^2 \;=\; 1.41 \times 10^{-17}\, \|f\|_2^2,$$

in both parity sectors separately and hence for arbitrary complex f.

(ii) (Upper bound.) *There is an explicit even test function f_{400}, a Legendre polynomial of degree 438 with rational coefficients of denominator 10^{40}, such that*

$$\lambda_Q(4/5) \;\le\; \frac{Q(f_{400})}{\|f_{400}\|_2^2} \;\le\; U \;=\; 1.90138291114546440603000 0 \times 10^{-17} \;<\; 1.9014 \times 10^{-17}.$$

(iii) (Parity separation.) *For every odd* $f \in L^2(-4/5, 4/5)$, $Q(f) \geq 2.3 \times 10^{-17}\|f\|_2^2$. *Since* f_{400} *is even and* $2.3 \times 10^{-17} - U > 3.98 \times 10^{-18} > 0$,

$$\inf_{f \text{ even}} \frac{Q(f)}{\|f\|_2^2} \;\leq\; U \;<\; 2.3 \times 10^{-17} \;\leq\; \inf_{f \text{ odd}} \frac{Q(f)}{\|f\|_2^2},$$

so the even infimum equals $\lambda_Q(4/5)$.

Parts (i) and (iii) are computer-assisted: they rest on interval arithmetic in the Arb midpoint–radius model [6], including all analytic error terms, on exact rational input data, and on a cold verifier that recomputes every decisive implication. Part (ii) needs one test function and is unconditional given the normalization.

Remark 3. Theorem 2(iii) identifies which sector carries the infimum. It does *not* assert that the infimum is attained, that a minimizer is unique, or that the least eigenvalue is simple. None of those is established here.

Remark 4. The bracket width is $1.9014 \times 10^{-17} - 1.41 \times 10^{-17} < 4.92 \times 10^{-18}$, about 35% of the upper endpoint. Section 6 measures where that width comes from and bounds how much of it the present method can remove.

3 The certified lower bound

The lower bound is assembled in five steps. Steps 1–5 are proved here in the normalization of Section 2; the strategy is that of [9, Thm. 1.1].

3.1 Step 1: the exterior gate

Lemma 5. $t \mapsto \operatorname{Re}\psi(\frac{1}{4} + \frac{it}{2})$ *is strictly increasing on* $t > 0$.

Proof. From the digamma series,

$$\operatorname{Re}\psi\Big(\tfrac{1}{4} + \tfrac{it}{2}\Big) = -\gamma + \sum_{n \geq 0}\left[\frac{1}{n+1} - \frac{n + 1/4}{(n+1/4)^2 + t^2/4}\right],$$

and each summand is strictly increasing in $t > 0$. Termwise differentiation is justified by local uniform convergence. □

Consequently, with $A_L = \sum_{\log n < 2L} 2\Lambda(n)/\sqrt{n}$ and $\beta := H(T) - A_L$, we get $\Psi_L(t) \geq \beta$ for all $|t| \geq T$, because $P_L \leq A_L$ pointwise. At $L = 4/5$,

$$A_L = 2.941973525223620455503909 \pm 3.91 \times 10^{-25},$$

and the gate values at the three cutoffs used in this note are

T	$\beta = H(T) - A_L$	status
200	$0.518465733243846719987353 6 \pm 2.62 \times 10^{-26}$	certified
400	$1.211613695058064549471502 \pm 1.89 \times 10^{-26}$	certified
800	$1.904761070930776888621206 \pm 1.68 \times 10^{-25}$	certified

with the increments $\beta_{400} - \beta_{200} = 0.693147961814217829484148 4 \pm 7.24 \times 10^{-27}$, numerically close to $\log 2$ as the asymptotics of H suggest. All three are *positive*, so the set $\{|t| \geq T : \Psi_L(t) < \beta\}$ is empty and no occupation or measure estimate on a bad exterior set is needed anywhere below.

3.2 Step 2: the comparison form

Because $\int_{|t|\ge T}(\Psi_L-\beta)|F|^2\ge 0$, equation (1) gives $Q(f)\ge R_{T,\beta}(f)$ with

$$R_{T,\beta}(f) \;=\; \beta\|f\|_2^2 \;+\; 2sF(i/2)^2 \;+\; \frac{1}{2\pi}\int_{-T}^{T}(\Psi_L(t)-\beta)|F(t)|^2\,dt. \tag{3}$$

The discarded quantity is exactly

$$e_T(f) \;:=\; \frac{Q(f)-R_{T,\beta}(f)}{\|f\|_2^2} \;=\; \frac{1}{2\pi\|f\|_2^2}\int_{|t|>T}(\Psi_L(t)-\beta)|F(t)|^2\,dt \;\ge\; 0, \tag{4}$$

which is measured in Section 6.

Proposition 6 (monotonicity in β)**.** *By Plancherel, $R_{T,\beta}(f)=2sF(i/2)^2+\frac{1}{2\pi}\int_{|t|\le T}\Psi_L|F|^2+\frac{\beta}{2\pi}\int_{|t|>T}|F|^2$, so for $\beta_2\ge\beta_1$,*

$$R_{T,\beta_2}(f)-R_{T,\beta_1}(f)=\frac{\beta_2-\beta_1}{2\pi}\int_{|t|>T}|F|^2 \;\ge\; 0.$$

On the finite compression this reads $A(\beta_2)-A(\beta_1)=(\beta_2-\beta_1)(I-K_T)$, where K_T is the compression of the Fourier energy-concentration operator and $0\le K_T\le I$.

Proposition 6 is recorded because it rules out a tempting move: *lowering* β cannot raise the exact comparison form, its infimum, or the least eigenvalue of its compression to a fixed subspace. Inspecting only the factor $\Psi_L-\beta$ inside the integral misses the $\beta\|f\|_2^2$ term. The identity was checked numerically with maximum residual 5.23×10^{-143} and diagonal entries of K_T in $[0.9420888637695,\ 0.9980132600038]$ at $T=200$, $N=16$.

3.3 Step 3: the Legendre basis

Let $\varphi_n(u)=\sqrt{(2n+1)/(2L)}\,P_n(u/L)$ be the orthonormal Legendre basis of $L^2(-L,L)$. Then

$$F_n(t) \;=\; i^n\sqrt{2L(2n+1)}\,j_n(Lt), \qquad \sum_{n\ge0}(2n+1)j_n(x)^2=1,$$

with j_n the spherical Bessel function. The form is assembled in the real coordinates $y_n=\sigma_n x_n$ with $\sigma_n=(-1)^{\lfloor n/2\rfloor}$, which makes $v_n(t)=\sqrt{2L(2n+1)}\,j_n(Lt)$ the real working transform. This basis is used, rather than the cosine basis of the repository's original assembly, because $j_n(x)$ is super-exponentially small for $n>ex/2$, which is what makes Step 4 effective; a cosine basis gives only $O(n^{-2})$ tail decay and cannot reach the required magnitude at an acceptable dimension.

3.4 Step 4: the complement

For $f\in V_N^{\perp}$ with $\|f\|_2=1$, Cauchy–Schwarz in n gives $|F(t)|^2\le 2L\,S_N(tL)$ with $S_N(x)=\sum_{n\ge N}(2n+1)j_n(x)^2$. Since S_N is increasing on $0\le x<N$,

$$g_N \;:=\; \sup_{f\in V_N^{\perp},\,\|f\|=1}\frac{1}{2\pi}\int_{-T}^{T}|F|^2 \;\le\; \frac{2LT}{\pi}\,S_N(TL).$$

This is a lossy but rigorous surrogate for the $(N+1)$-st Slepian eigenvalue [7], and it is super-exponentially small past the Shannon number $2TL/\pi$. At $T=400$, $N=440$:

$$S_N(320)=9.260051461963608\times10^{-63}, \qquad g_N=1.886442193224756\times10^{-60}.$$

3.5 Step 5: the Schur-type block bound, with strict tests

Split $f = f_N + f_\perp$. With a a certified lower bound for the least eigenvalue of the compression of $R_{T,\beta}$ to V_N, d a certified lower bound for $R_{T,\beta}$ on $V_N^\perp$, and κ a certified bound on the coupling, a 2×2 comparison yields positivity as soon as $ad > \kappa^2$. The threshold κ^2/d is a *break-even point*, not the resulting bound. To avoid the cancellation-prone explicit eigenvalue formula, the final implication is tested as three *strict* interval signs for the proposed constant μ:

$$a > \mu, \qquad d > \mu, \qquad (a-\mu)(d-\mu) > \kappa^2. \tag{5}$$

Equality anywhere in (5) would give only a nonnegative comparison, so strictness is required and is checked as a sign of an interval, never of a midpoint. The coupling constant is $\kappa = 2|p_N||p_\perp| + M\sqrt{g_N}$ with $M = \sup_{|t|\le T} |\Psi_L - \beta|$ and p the pole vector $p_n = \sqrt{2L(2n+1)}\, i_n(L/2)$ taken in the same y-coordinates; $|p|^2 = \int_{-L}^{L} e^{-u}\, du = 2\sinh L$ exactly. M is obtained from the monotonicity of H alone, with no covering argument.

3.6 Verified positive definiteness

A naive interval LDL^{T} factorization is unusable at this dimension: entry radii of order 10^{-38} amplify by roughly 6.5 per pivot and reach $O(1)$ by step 47. Instead, positive definiteness of $B = A - aI$ is established by a preconditioned congruence in the sense of Rump [5]. Let R be the upper triangular Cholesky factor of the *midpoint* of B, treated as exact input, and set $C = R^{-1}$ and

$$G \;=\; C^* B\, C \;=\; R^{-\mathsf{T}}\,(A - aI)\,R^{-1},$$

computed by two explicit forward substitutions in ball arithmetic. The verified inequality is

$$\|I - G\|_\infty \;<\; 1.$$

This proves $B \succ 0$: the bound places every eigenvalue of the Hermitian G in $(0, 2)$, so $G \succ 0$; if C were singular then $G = C^*BC$ would be singular, so C is nonsingular; and then for $x \neq 0$, writing $x = Cy$ with $y \neq 0$, $x^*Bx = y^*Gy > 0$. It is a congruence, not a one-sided inverse residual — the latter would establish nonsingularity only.

3.7 Quadrature

The interior integral in (3) is evaluated by panelled Gauss–Legendre quadrature with certified nodes and weights (FLINT's `legendre_p_root`). The symbol, prime comb and the full basis-transform vector are evaluated once per node and reused across all matrix entries, so no entry is integrated adaptively. The quadrature remainder is bounded by a Bernstein-ellipse estimate on genuinely holomorphic extensions: the half-sum

$$H_{\mathrm{hol}}(z) \;=\; \tfrac{1}{2}\Big[\psi\Big(\tfrac{1}{4} + \tfrac{iz}{2}\Big) + \psi\Big(\tfrac{1}{4} - \tfrac{iz}{2}\Big)\Big] - \log\pi$$

agrees with H on the real axis and has its nearest poles at $z = \pm i/2$ (note that $\operatorname{Re}\psi(\frac{1}{4} + \frac{iz}{2})$ is *not* a holomorphic extension), and the basis product is extended as $v_n(z)v_m(-z)$ rather than by conjugation at a complex argument. On the ellipses, $|v_n(z)|$ is bounded uniformly in n by

$$\Big|\int_{-L}^{L} \varphi(u) e^{izu}\, du\Big| \;\le\; \Big(\int_{-L}^{L} e^{-2u\operatorname{Im} z} du\Big)^{1/2} \;\le\; \sqrt{2L}\, e^{L|\operatorname{Im} z|} \tag{6}$$

for any L^2-normalized φ supported in $[-L, L]$. Inequality (6) is immediate from Cauchy–Schwarz and is *not* claimed as new; it is recorded because it replaces a direct interval evaluation of j_n on wide complex balls, which is numerically useless here. The remainder is folded into the matrix entries as a widening exactly once.

3.8 The $T = 400$ certificate

parameter	value
window	$L = 4/5$
cutoff	$T = 400$
dimension	$N = 440$ (Legendre degrees 0..439; 220 modes per parity)
quadrature	800 panels of half-length 1/4, Gauss–Legendre order 64
Bernstein data	$b_e = 0.30$ (inside the strip $\lvert\operatorname{Im} z\rvert < 1/2$), $\rho = 2.762049935$
remainder factor	$3.551349410 \times 10^{-57}$
precision	900 bits (Bessel), 384 bits (accumulation), 1400 bits (verification)

$N = 440$ was chosen from the actual inequalities, not from the heuristic $eLT/2 = 434.9$. The certified constants are

$$\beta = 1.21161369505806454947150 2, \qquad M = 9.525770639507350587208369,$$
$$\kappa = 1.30834377128538825902901 8 \times 10^{-29}, \qquad \kappa^2/d = 1.412796364751588 \times 10^{-58},$$

with pole tail $|p_\perp| = 7.993372152149564 \times 10^{-1282}$.

sector	a (certified)	$\|I - G\|_\infty$	d	μ	margin in (5)
even	1.42×10^{-17}	$1.4973903 \times 10^{-14}$	1.211613695058065	1.41×10^{-17}	$1.21161369506 \times 10^{-19}$
odd	2.4×10^{-17}	$4.3535838 \times 10^{-22}$	1.211613695058065	2.3×10^{-17}	$1.21161369506 \times 10^{-18}$

The whole-window constant is the smaller of the two sector constants, so $\mu = 1.41 \times 10^{-17}$ and *the even sector binds*. Assembly took 3532 s, the full run 3581 s, with peak resident memory 393.4 MiB; the two Rump factorizations at $N = 440$ took under 50 s together.

3.9 Parity and complex tests

The form decouples by parity: the interior matrix has $C_{nm} = 0$ when $n + m$ is odd, because $v_n v_m$ is then odd on a symmetric interval, and the pole term $2(C^2 - S^2)$ separates into its even and odd parts. For complex $f = u + iv$ the Hermitian form gives $Q(f) = Q(u) + Q(v)$, and each real part splits by parity. Hence the two sector certificates cover the full class, which is what makes Theorem 2(i) a statement about arbitrary complex f.

Remark 7. A positive-definiteness check cannot serve as an oracle for the pole sign. The direct convention checks of Section 4 are retained for that purpose, because an earlier version of this computation carried a pole-phase error that produced a plausibly negative pivot (Appendix A, item 1).

4 The original-form upper bound

4.1 Why the time domain

A variational upper bound requires Q itself, not $R_{T,\beta}$. Evaluating (1) directly would mean certifying an oscillatory integral over all of $\mathbb{R}$. Instead Q is evaluated on the geometric side via the autocorrelation

$$g(x) \;=\; \int f(s + x) f(s)\, ds,$$

supported in $[-2L, 2L]$, together with the compact archimedean identity ([9, Lemma 2.5], transcribed into the present normalization)

$$\operatorname{Arch}(g) = -\big[\gamma_E + \log\pi + \log(1 - e^{-4L})\big] g(0) + \int_0^{2L} \frac{2\big[e^{-2x} g(0) - e^{-x/2} g(x)\big]}{1 - e^{-2x}}\, dx, \qquad (7)$$

$$Q(f) = 2sF(i/2)^2 + \mathrm{Arch}(g) - \sum_{\log n < 2L} \frac{2\Lambda(n)}{\sqrt{n}}\, g(\log n). \tag{8}$$

This is the route by which [9, §6.2] evaluates its own variational upper bounds at this window, there with shift-correlation integrands of degree at most 239 integrated exactly by a 160-node Gauss rule. Two features make it cheap and rigorous. They are set out below because the present error accounting depends on them, not as new observations.

Exact inner quadrature. If f is a polynomial of degree d on $[-L, L]$ then g is a polynomial of degree at most $2d$ on $[0, 2L]$, and the integrand $f(s+x)f(s)$ has degree at most $2d$ in the integration variable, so an m-node Gauss–Legendre rule is exact once $2m-1 \geq 2d$, i.e. $m \geq d+1$. The implementation uses $m = d+2$; for $d = 438$ that is 440 nodes. *Exactness* here means no polynomial quadrature truncation error in exact arithmetic — node, weight, coefficient and rounding enclosures are still carried as ball radii, and the outer integral in (7) keeps its own certified Bernstein remainder.

The removable endpoint, by an identity. The integrand of (7) is removable at $x = 0$ and its limit involves $c = -g'(0^+)$. For even f,

$$g'(x) \;=\; -f(L)f(L-x) + \int_{-L}^{L-x} f'(s+x)f(s)\, ds,$$

and at $x = 0$ the integral is $\frac{1}{2}[f^2]_{-L}^{L} = 0$, so

$$c \;=\; -g'(0^+) \;=\; f(L)^2 \qquad \text{exactly,} \tag{9}$$

and the integrand limit is $c - \frac{3}{2}g(0)$. This closed form is an implementation convenience obtained by elementary differentiation and integration; [9, Lemma 2.5] leaves c as the one-sided limit and remarks only that it is nonzero in general. Its absence from the displayed formulas there is not evidence that it is new, and no novelty is claimed for it. The outer integrand is analytic in $|\operatorname{Im} x| < \pi$; panels are geometric near the origin so that no Bernstein ellipse encloses it, and the residual head $[0, 10^{-40}]$ is bounded by an explicit constant.

4.2 Validation of the representation

(a) On the constant $f = \varphi_0$, the compact route gives $Q(\varphi_0) = 0.080274054959269611214 \pm 3.09\times10^{-22}$, matching the repository's independent closed form $Q[0,0] = 0.080274054959269611214$ to all 21 printed digits (difference -3.08×10^{-22}). That closed form is the evaluator of [10], which sums the archimedean series analytically through the digamma, Lerch Φ and Hurwitz ζ functions; it is an independent route to Q, not an independent implementation of the present one.

(b) A degree-0 check cannot see a pole-phase error, since $\sigma_0 = +1$. Checks at φ_2 and at the signed mixed function $\varphi_0 - \varphi_2$ were therefore added, with the pole moment computed *directly* from the physical-space function rather than from the pole-vector helper, and with the finite prime shifts re-enumerated independently of the symbol helper:

f	$Q(f)$	pole moment (direct)
φ_0	0.080274054959269611214	1.29891290374844725
φ_2	0.389336213205013655883	0.030516224225381024
$\varphi_0 - \varphi_2$	0.117094370226589934780	1.26839667952306623

All agree with the helper values, the first to 10^{-25}.

(c) *Disclosure.* This is a second *representation* of the same form, not an independent implementation: it shares the Arb library, the symbol helper for the comb data, and the pole-vector helper. It is not described as an independent check.

4.3 The frozen test functions and the bound

Two trial functions are used, each a frozen exact rational vector in the Legendre y-coordinates with denominator 10^{40}:

- f_{200}: 110 even modes, degrees 0..218, $\|f\|_2^2 = 2.27524444365096536097055 0$;
- f_{400}: 220 even modes, degrees 0..438.

Both were *located* by an exploratory floating-point eigensolver, then rounded to exact rationals and certified; the eigensolver contributes no certified claim, and in particular supplies no eigenvalue.

For f_{200}, equation (8) gives, at 94 outer panels and head cut 10^{-40},

$$\text{pole} = 3.2303213242823875400, \qquad \text{Arch} = -3.1402989129419417722,$$

$$\text{comb} = 0.090022411340445684793,$$

whose signed sum is $8.30143811294573721 \times 10^{-17}$: a cancellation of about sixteen digits, which is why the evaluation is done at high precision in ball arithmetic and why the compact representation — with its exact inner quadrature and the exact identity (9) — matters. Dividing by the norm,

$$q(f_{200}) = 3.64859175290407702 \times 10^{-17} \pm 4.15 \times 10^{-35}.$$

This is above the then-current upper bound and so does not improve it; the expected outcome, since f_{200} minimizes a comparison form, not Q.

For f_{400},

$$q(f_{400}) = 1.90138291114546440 \times 10^{-17} \pm 6.03 \times 10^{-35},$$

whose certified upper endpoint is

$$U \;=\; 1.901382911145464406030000 \times 10^{-17},$$

giving Theorem 2(ii).

Remark 8 (on quoting this number). The decimal $1.90138291114546440 \times 10^{-17}$ is the *midpoint* of the enclosure and is *not* a valid strict upper bound for it; the strict upper endpoint is U above. Earlier internal drafts of this work quoted the midpoint as the bracket's upper endpoint, and a separate internal record quoted a looser endpoint $1.9013829111454646892 \times 10^{-17}$ arising from a low-precision re-parse of the serialized ball; both are corrected (Appendix A, item 6). The rounded value 1.9014×10^{-17} used in the abstract and in Theorem 2 is above every one of them and is the safe form in which to quote the result.

5 The parity separation

The $T = 400$ certificate of Section 3 delivers $a = 2.4 \times 10^{-17}$ in the odd sector against $a = 1.42 \times 10^{-17}$ in the even sector. Re-evaluating the three strict signs (5) from the exact certified data — not from rounded report values — at $\mu_{\text{odd}} = 2.3 \times 10^{-17}$ gives

$$a > \mu_{\text{odd}}, \qquad d > \mu_{\text{odd}}, \qquad (a - \mu_{\text{odd}})(d - \mu_{\text{odd}}) > \kappa^2,$$

with margin $1.21161369506 \times 10^{-18}$. So the whole odd sector satisfies $Q(f) \geq 2.3 \times 10^{-17}\|f\|_2^2$. Combining with U from Section 4, and using that f_{400} is even,

$$\mu_{\text{odd}} - U \;=\; 3.98617088854535593970000 0 \times 10^{-18} \;>\; 0,$$

so

$$\inf_{\text{even}} \frac{Q(f)}{\|f\|_2^2} \;\leq\; U \;<\; \mu_{\text{odd}} \;\leq\; \inf_{\text{odd}} \frac{Q(f)}{\|f\|_2^2},$$

which with the orthogonal parity decomposition identifies the even infimum as $\lambda_Q(4/5)$. The global lower bound remains 1.41×10^{-17}, governed by the even sector. As noted in Section 1.3, none of this structure is new: at this very window [9, Thm. 6.2] certifies the same ordering with $\lambda_1^{\text{odd}} \geq 8.206 \times 10^{-15}$, more than two orders of magnitude above the constant obtained here. What is recorded above is only that this pipeline's own certified numbers separate the two sectors. Nothing about simplicity, uniqueness or attainment follows from them — simplicity and evenness at this window were previously established by [9, Thm. 6.2].

6 Cutoff diagnostics

6.1 The cutoff comparison, proved

Proposition 9. *Let $\Psi_L \geq \beta_i$ on $|t| > T_i$ for $i = 1, 2$, with $T_2 \geq T_1$ and $\beta_2 \geq \beta_1$. Then*

$$R_{T_2,\beta_2}(f) - R_{T_1,\beta_1}(f) = \frac{1}{2\pi} \int_{T_1<|t|\leq T_2} (\Psi_L - \beta_1)|F|^2 + \frac{\beta_2 - \beta_1}{2\pi} \int_{|t|>T_2} |F|^2 \;\geq\; 0,$$

both terms being nonnegative. Hence $\lambda_R(T_2) \geq \lambda_R(T_1)$ for the exact forms.

Two cautions. This does not imply that a *certified numerical* bound improves, since error terms change with the parameters. And it does not order finite matrices of different sizes: changing N changes the compression, so only matrices in the same subspace are comparable by this identity.

6.2 The scalar table

Each entry below is a certified enclosure on a *fixed, frozen* function; no reoptimization is involved, so the columns compare exact forms on the same vector.

	f_{200}	f_{400}
R_{200}	$1.0310220302459896 \times 10^{-17}$	$1.24275215 \times 10^{-17}$
R_{400}	$2.3520530467310079 \times 10^{-17}$	$1.4330724047355850 \times 10^{-17}$
R_{800}	3.01603×10^{-17}	1.6681×10^{-17}
Q	$3.6485917529040770 \times 10^{-17}$	$1.9013829111454644 \times 10^{-17}$

The R_{800} entries carry wider enclosures ($\pm 5.30 \times 10^{-23}$ and $\pm 4.60 \times 10^{-22}$) because the quadrature remainder grows with the cutoff; $R_{200}(f_{400})$ was evaluated as a direct scalar integral on the full degree-438 function, with no truncation and no enlarged matrix. Each column is strictly increasing in T, as Proposition 9 requires.

6.3 What the comparison discards

From (4), with $e_T(f) = q(f) - r_T(f)$ by rigorous interval subtraction:

	f_{200}	f_{400}
e_{200}	$2.617569722658087 \times 10^{-17}$	6.586308×10^{-18}
e_{400}	$1.296538706173069 \times 10^{-17}$	$4.683105064098794 \times 10^{-18}$
e_{800}	6.3256×10^{-18}	2.333×10^{-18}
e_{400}/e_{200}	$0.49532156 \pm 4.69 \times 10^{-9}$	$0.7110365 \pm 1.61 \times 10^{-8}$
e_{800}/e_{400}	$0.48788 \pm 9.07 \times 10^{-6}$	$0.498 \pm 2.36 \times 10^{-4}$

There is no decay law here. On f_{400} the successive ratios are 0.711 and then 0.498, which is not a constant rate; and the two functions disagree at the *same* doubling $200 \to 400$, giving 0.495 and 0.711, so the ratio is not a function of T alone. That both functions give approximately 0.49 across $400 \to 800$ is two samples of one step, not a rate in T, and comparing differently optimized vectors would establish less still. No uniform operator estimate is inferred from a fixed-vector ratio, and none of these ratios is used in any certified statement.

6.4 What $q_f - r_f$ is, and is not

For f_{400}, $q_f - r_f = 4.6831050640988 \times 10^{-18}$ is the discarded-term identity (4) evaluated on that function, inside its certified enclosure. It is *not* $\lambda_Q - \lambda_R(400)$: both $\lambda_R(400) \le r_f$ and $\lambda_Q \le q_f$ are upper bounds, and two upper bounds do not make their difference an equality between infima.

6.5 Certified ceilings, and the stopping decision

Because f_{400} and f_{200} are admissible, each R_T value in the table is a variational *upper* bound for $\lambda_R(T)$, and any certificate resting only on $Q \ge R_{T,\beta}$ is capped by it. Relative to the lower bound in force:

route	certified ceiling	reference lower bound	headroom
R_{200}	$\lambda_R(200) \le r_{200}(f_{200}) = 1.0310220 \times 10^{-17}$	1×10^{-17}	3.102%
R_{400}	$\lambda_R(400) \le r_{400}(f_{400}) = 1.4330724 \times 10^{-17}$	1.41×10^{-17}	1.636%
R_{800}	$\lambda_R(800) \le r_{800}(f_{400}) = 1.6681 \times 10^{-17}$	1.41×10^{-17}	18.31%

So the $T = 400$ route is within about 1.6% of exhausted, and the entire $T = 800$ route is capped at a factor 1.1830 over the present lower endpoint; since a reoptimized f_{800} can only sit below that ceiling, the realizable gain is strictly less. Measured costs on the machine used:

stage	measurement
assembly, $T = 200$, $N = 220$, 800×64 nodes	1338 s, peak 205.5 MiB
assembly, $T = 400$, $N = 440$, same node count	3532 s, peak 393.4 MiB
Rump factorizations, both sectors, $N = 440$	under 50 s
compact $Q(f_{200})$	389 s
cross table (R_{400}, $Q(f_{400})$)	1544 s
scalar leg $R_{200}(f_{400})$, 12800 nodes	933 s
scalar leg $R_{800}(f_{400})$, 51200 nodes	8049 s
scalar leg $R_{800}(f_{200})$, 51200 nodes	3651 s

The assembly grew by a factor 2.64 when N doubled at fixed node count, and the $T = 800$ scalar legs overran the node-count scaling by more than a factor of two, which indicates that per-Bessel cost grows with the argument rather than node count alone. Extrapolating — as an estimate anchored on these measurements, not a guarantee — a full $T = 800$ matrix at $N \approx 880$ with roughly 1600×64 nodes would cost on the order of ten hours of assembly. Against a ceiling

of +18.31%, the recommendation recorded in the accompanying repository is to **consolidate at** $T = 400$. Neither the pilot nor any failed certificate here is an intrinsic barrier; this is a cost–benefit judgement from measured data, and a separately scoped $T = 800$ attempt would be a reasonable thing to run with that ceiling stated in advance.

7 Reproducibility and limitations

7.1 Artifacts and verification

Every decisive quantity in this note is published as a midpoint–radius enclosure in machine-readable form, together with the exact rational input vectors and the code that produced them. Printed decimals in this note are summaries; the enclosures are authoritative. Six independent verifiers recompute the chain from the frozen artifacts and exit nonzero on any failure; on the run accompanying this version all six exit 0 with 119 checks passing in total $(34 + 19 + 20 + 16 + 18 + 12)$, in 97 seconds. A seventh verifier, shipped with this note, re-derives every decimal quoted above from the artifacts and additionally checks each quoted bound's *rounding direction* and its scope and parity label (47 further checks); it is what caught the mis-quoted upper endpoint of Appendix A, item 6.

The verifiers fail closed. Any NaN, infinite radius, indeterminate enclosure, or unresolved sign produces an explicit failure or a refinement request; no skipped cell, panel or pivot is interpreted as success. This matters because an earlier version of the computation silently skipped cells where a library digamma call returned NaN on wide balls and thereby produced a wrong constant that was *used* (Appendix A, item 2).

7.2 Environment

Linux, Python 3.14.7, `mpmath 1.3.0`, `python-flint 0.9.0` (FLINT/Arb). Floating-point `mpmath` is used only to *locate* candidate vectors, which are then frozen as exact rationals and certified; it appears in no pass/fail decision. Arb working precisions on sign-critical paths range from 256 to 2000 bits and are stated per stage in the artifacts.

7.3 Limitations

1. **One window.** Everything is at $L = 4/5$. There is no propagation theorem, and nothing follows at any larger support. The repository's companion measurement at $L = 81/100$ indicates roughly $75\times$ the assembly work for a target that is certifiably at most 6.565×10^{-18}, which is why that window was not attempted here.

2. **An imported normalization.** The explicit-formula pairing in the convention of Section 2 is imported from the accompanying repository, not rederived. It was cross-checked only through the independent closed form for $Q[0, 0]$ and the degree-2 convention checks of Section 4.

3. **An unverified external matrix step.** Zhu describes supplementary source code and certificates in [9, §17]; these were not obtained or executed in the present work, so the matrix step of [9, Thm. 1.1] is not reproduced here. The present certificate is independent of it.

4. **Shared arithmetic.** The compact time-domain evaluator shares library code with the matrix assembly, and is reported as a second representation, not an independent implementation.

5. **No arithmetic content.** The only analytic sharpening proved here is the monotone gate of Lemma 5, which moves a constant by under one percent and does not touch the doubly exponential threshold that governs the cost in L.

6. **Scope.** No zero-free region, no verified range of zeros, no statement about the Riemann hypothesis.

7.4 Revision history

1.0.0 (2026-11-10) First version.

1.0.1 (2026-11-10) Citation and provenance revision only. Adds the two earlier technical notes whose implementation and parity conventions this computation inherits ([10] and [11]), the lineage paragraph in Section 1.3, and the corresponding citations in Sections 2 and 4. No numerical result, bound, artifact or limitation changed; the certified bracket and the parity statement are exactly those of version 1.0.0.

1.0.2 (2026-11-10) Corrections after reading [9] in full, which had previously been consulted only through its abstract and rendered HTML. Three attributions were too weak and are strengthened: [9, Thm. 6.2] certifies the parity *separation* at this very window, with a far stronger odd-sector constant, and establishes simplicity and evenness of the ground state (Sections 1.3 and 5); [9, §6.2] already evaluates its variational upper bounds by the time-domain route used in Section 4; and the supplementary materials of [9, §17] are described there, rather than absent. Section 1.4 is new and compares the two sets of bounds at this window. Again no numerical result, bound or artifact changed.

A Error history

Errors found during this work are recorded because several of them produced plausible wrong numbers rather than failures, and because the countermeasures are part of what the artifacts are for. Items 1–5 are library- or discipline-level defects; items 6–8 are corrections to claims. A longer ledger, including retracted statements $R1$–$R13$, is kept with the artifacts.

1. **Pole-coordinate phase.** The form is assembled in the coordinates $y_n = \sigma_n x_n$ with $\sigma_n = (-1)^{\lfloor n/2 \rfloor}$, so the pole vector requires the same phase. Omitting it is invisible at degree 0 and wrong from degree 2 onward. It produced a genuinely negative pivot (-0.3949) that would have been reported as indefiniteness. Caught by comparing $|F(t)|^2$ and $F(i/2)$ against direct integration; the degree-2 checks of Section 4 now cover it.

2. **Digamma on wide balls.** The library digamma returns NaN on balls whose real part has radius $\gtrsim 0.1$, even strictly inside the right half-plane. A comparison loop silently skipped those cells and returned $M \approx 1831$ instead of 8.83 — and the wrong value was used. Replaced by the explicit recurrence $\psi(z) = \psi(z+K) - \sum_{k<K}(z+k)^{-1}$ with adaptive K, and by a covering-free monotonicity bound for M.

3. **Bessel functions on wide balls.** Interval evaluation of j_n on the complex ellipses is useless here (width $\sim 10^{32}$ at $|w| \sim 80$, unchanged at 3000 bits). Replaced by the uniform Cauchy–Schwarz bound (6).

4. **Interval factorization.** Naive interval LDL^{T} is unusable at these dimensions (radii amplify about $6.5\times$ per pivot, reaching $O(1)$ by step 47). Replaced by the Rump preconditioned congruence.

5. **Precision ownership.** The Arb context precision is a process global. Routines set it on entry and did not restore it, so a callee silently changed its caller's context; this produced four separate wrong constants in this project, each plausible-looking rather than an error (a complement constant at $[\pm 7.4 \times 10^{132}]$, an even-sector d at $[\pm 2.8 \times 10^{183}]$, and two coefficient sets built at 53 bits). The repair is an owned-scope wrapper that asserts restoration on normal return *and* on exception, caches keyed by precision so a low-precision value can never be served to a high-precision caller, and an explicit *unresolved* result for any wide or non-finite enclosure. The lesson, and the harder half, is that *an input constructed outside the scope carries the ambient precision's radius, and entering a higher-precision scope afterwards does not recover it*: own your inputs as well as your scope.

6. **The upper endpoint, twice mis-quoted.** Two distinct errors, both in the safe direction numerically but both wrong as stated. (a) Internal drafts quoted the enclosure *midpoint* $1.90138291114546440 \times 10^{-17}$ as the bracket's upper endpoint; a midpoint is not an upper bound for its own ball. (b) A separate record quoted $1.9013829111454646892 \times 10^{-17}$ as "the exact upper endpoint"; that value came from re-parsing the serialized ball at the ambient 53 bits, which rounds the midpoint and then inflates the endpoint by about 2.8×10^{-33} — the precision-ownership defect of item 5 recurring in the very routine that documents it. The strict endpoint is $U = 1.90138291114546440603000 \times 10^{-17}$, and 1.9014×10^{-17} is the safe rounded form. See Remark 8.

7. **A retracted sparsity target.** An earlier proposed next step — a power-saving sparsity bound on the near-resonant set of the prime comb — is false, and is retracted in full: by continuous Kronecker–Weyl equidistribution that set has positive density, confirmed numerically here to four significant figures across four decades of T. The replacement is the empty-exterior gate of Section 3, which needs no such estimate.

8. **Overstated interpretations, corrected.** Four further claims made alongside earlier versions were withdrawn or corrected: that no bound on the measure of a bad set could control $\int_E |F|^2$ (false — standard finite-measure and local-occupation estimates do exactly that); that a Schur *threshold* was a lower bound (it is a break-even point, and the final test needs strict inequalities, hence (5)); that lowering β would buy a larger a (refuted by Proposition 6 — the $\beta\|f\|_2^2$ term had been ignored); and an accusation that the library's adaptive integrator returned a silently wrong value (withdrawn — it could not be reproduced; the wrong numbers came from a run that still carried the pole-phase error of item 1, at a different cutoff, with output from two runs conflated).

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